(c) The production cost of the company during the period 2012- 2014 is less than 40 lakhs.

(iii) Average production Cost of the company

$$= \frac{55 + 40 + 30 + 25 + 35 + 70}{6}$$

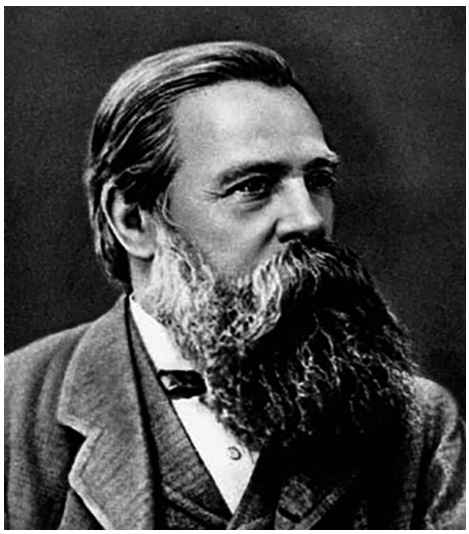
$$= ₹ 42.5 \text{ Lakhs}$$

(iv) Percentage increase in the production cost of the company is

$$= \frac{70 - 35}{35} \times 100$$

$$= 100\%$$

4.3.2 Pareto Diagram:



Vilfredo Pareto (1848-1923), born in Paris in an Italian aristocratic family, studied Engineering and Mathematics at the University of Turin. During his studies at the University of Lousane in Switzerland, Pareto derived a complicated mathematical formula to prove the distribution of income and wealth in society is not random. Approximately 80% of total wealth in a society lies with only 20% of the families. The famous law about the 'Vital few and trivial many' is widely known as 'Pareto Principle' in Economics.

Pareto diagram is similar to simple bar diagram. But, in Pareto diagram, the bars are arranged in the descending order of the heights of the bars. In addition, there will be a line representing the cumulative frequencies (in %) of the different categories of the variable. The line is more useful to find the vital categories among trivial categories

Example 4.2

Administration of a school wished to initiate suitable preventive measures against breakage of equipment in its Chemistry laboratory. Information collected about breakage of equipment occurred during the year 2017 in the laboratory are given below: